

EE/CE 252/251

Signals and Systems

with

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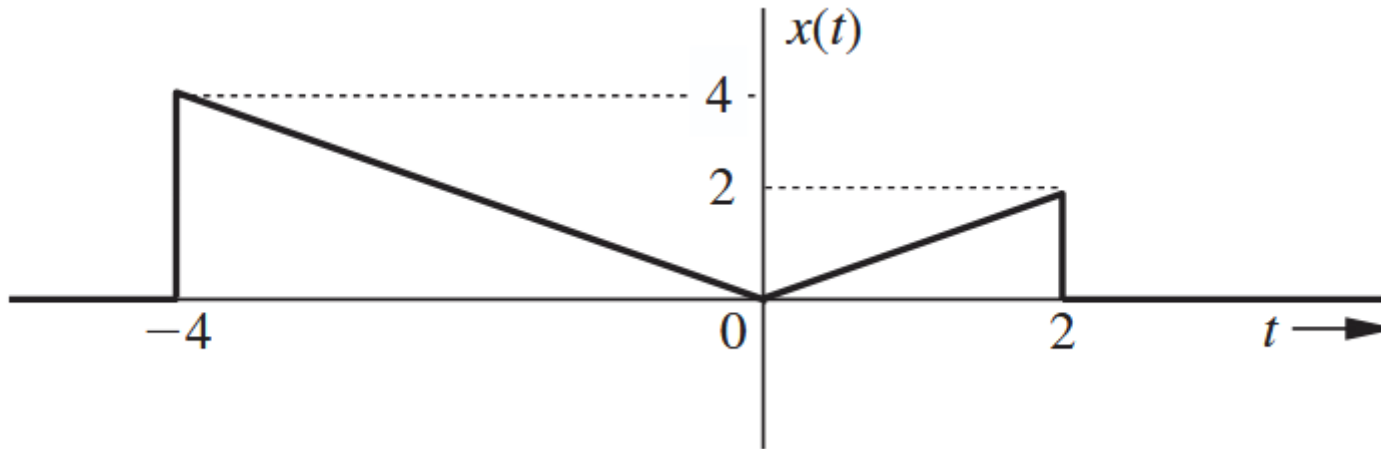
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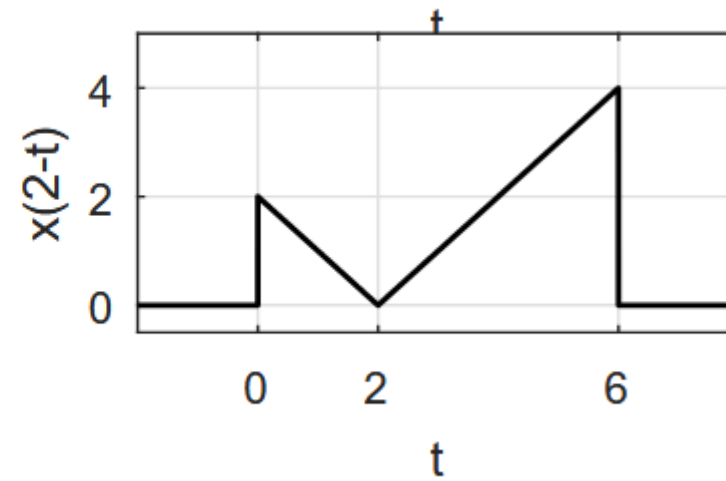
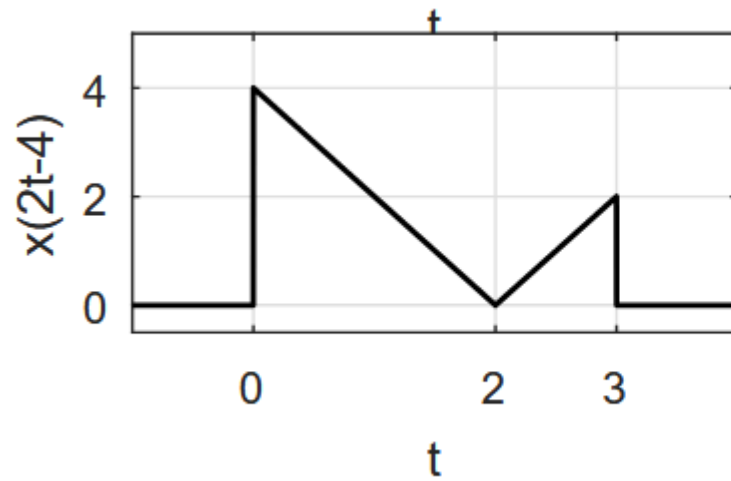
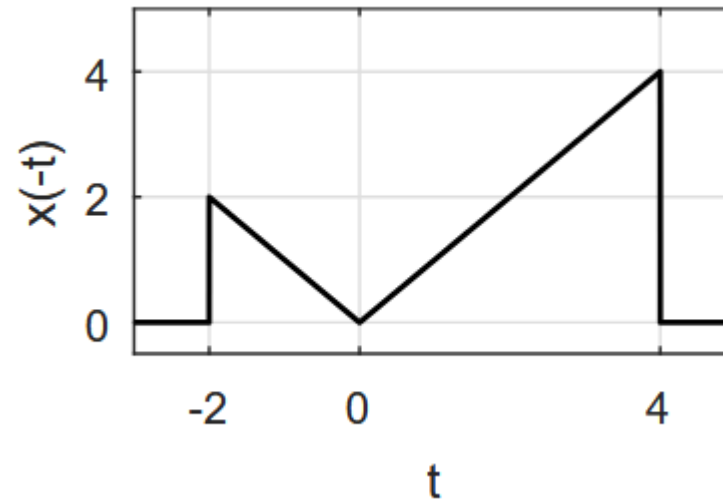
Problem 1

For the signal shown plot the following

- $x(-t)$
- $x(2t - 4)$
- $x(2 - t)$

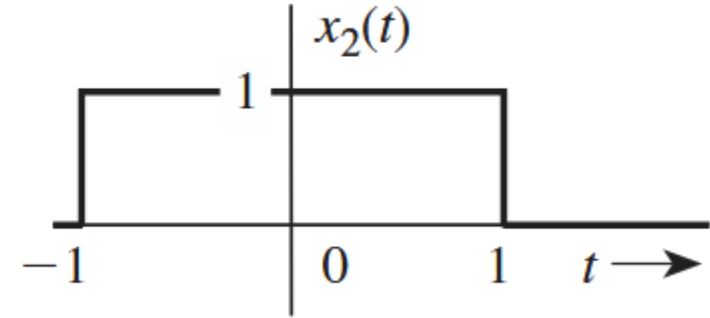
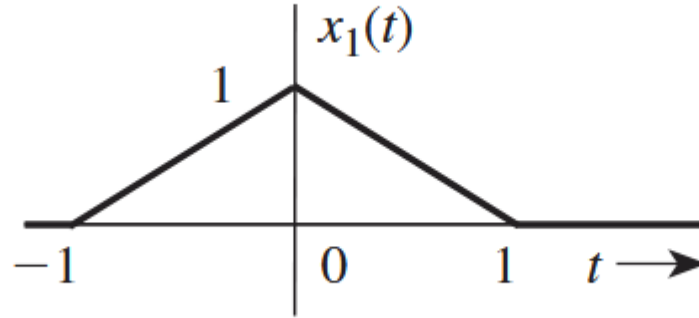
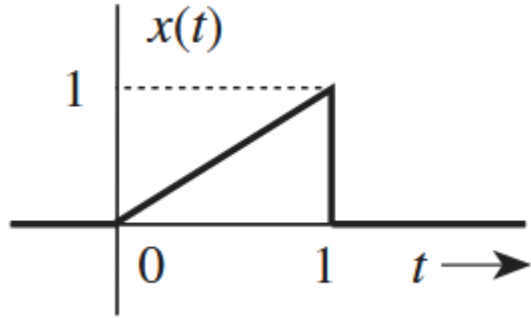


Solution



Problem 2

Write x_1, x_2 in terms of x



Solution

- (a) $x_1(t)$ can be formed by shifting $x(t)$ to the left by 1 plus a time-inverted version of $x(t)$ shifted to left by 1. Thus,

$$x_1(t) = x(t + 1) + x(-t + 1) = x(t + 1) + x(1 - t).$$

- (b) $x_2(t)$ can be formed by time-expanding $x(t)$ by factor 2 to obtain $x(t/2)$ Now, left-shift $x(t/2)$ by unity to obtain $x(\frac{t+1}{2})$. We now add to this a time-inverted version of $x(\frac{1-t}{2})$ to obtain $x_2(t)$. Thus,

$$x_2(t) = x(\frac{t+1}{2}) + x(\frac{1-t}{2}).$$

Problem 3

Consider an energy signal $x(t)$. Which of the following operations will not change its energy?

1. Time-shifting
2. Time-reversal
3. Time-compression
4. Time-expansion
5. Amplitude-scaling

Justify your claims.

Solution

$$E_{-x} = \int_{-\infty}^{\infty} [-x(t)]^2 dt = \int_{-\infty}^{\infty} x^2(t) dt = E_x$$

$$E_{x(-t)} = \int_{-\infty}^{\infty} [x(-t)]^2 dt = \int_{-\infty}^{\infty} x^2(x) dx = E_x$$

$$E_{x(t-T)} = \int_{-\infty}^{\infty} [x(t-T)]^2 dt = \int_{-\infty}^{\infty} x^2(x) dx = E_x,$$

$$E_{x(at)} = \int_{-\infty}^{\infty} [x(at)]^2 dt = \frac{1}{a} \int_{-\infty}^{\infty} x^2(x) dx = E_x/a$$

$$E_{x(at-b)} = \int_{-\infty}^{\infty} [x(at-b)]^2 dt = \frac{1}{a} \int_{-\infty}^{\infty} x^2(x) dx = E_x/a,$$

$$E_{x(t/a)} = \int_{-\infty}^{\infty} [x(t/a)]^2 dt = a \int_{-\infty}^{\infty} x^2(x) dt = aE_x$$

$$E_{ax(t)} = \int_{-\infty}^{\infty} [ax(t)]^2 dt = a^2 \int_{-\infty}^{\infty} x^2(t) dt = a^2 E_x$$

Problem 4

If $x_e(t)$ and $x_o(t)$ are even and the odd components of a real signal $x(t)$, then show that

$$\int_{-\infty}^{\infty} x_e(t)x_o(t) dt = 0$$

Solution

$$\begin{aligned}x_e(t)x_o(t) &= \frac{1}{4}[x(t) + x(-t)][x(t) - x(-t)] \\&= \frac{1}{4}[|x(t)|^2 - |x(-t)|^2]\end{aligned}$$

Since the areas under $|x(t)|^2$ and $|x(-t)|^2$ are identical, it follows that

$$\int_{-\infty}^{\infty} x_e(t)x_o(t)dt = 0$$

Problem 5

Find and sketch the odd and the even components of the following:

(b) $tu(t)$

(c) $\sin \omega_0 t$

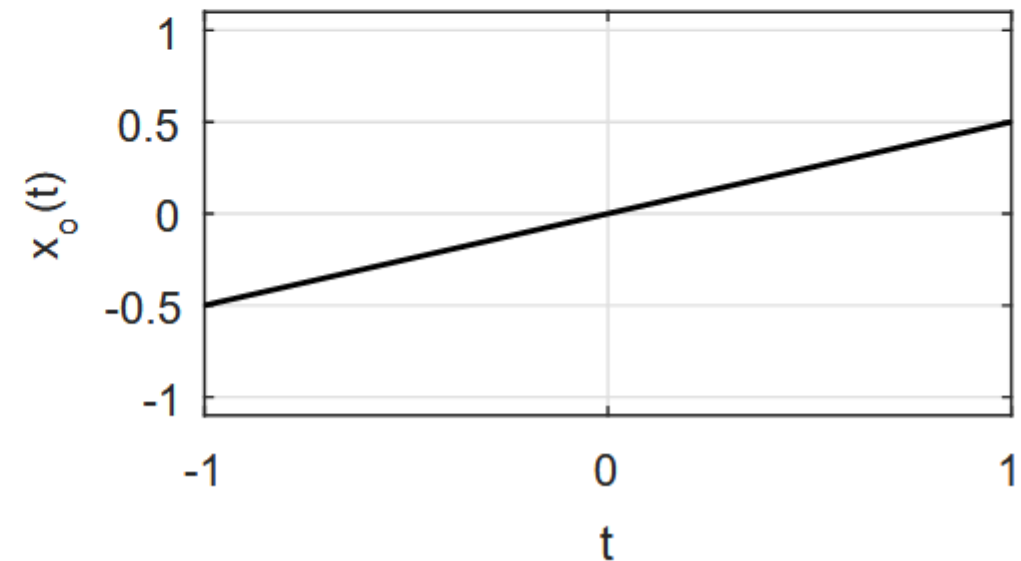
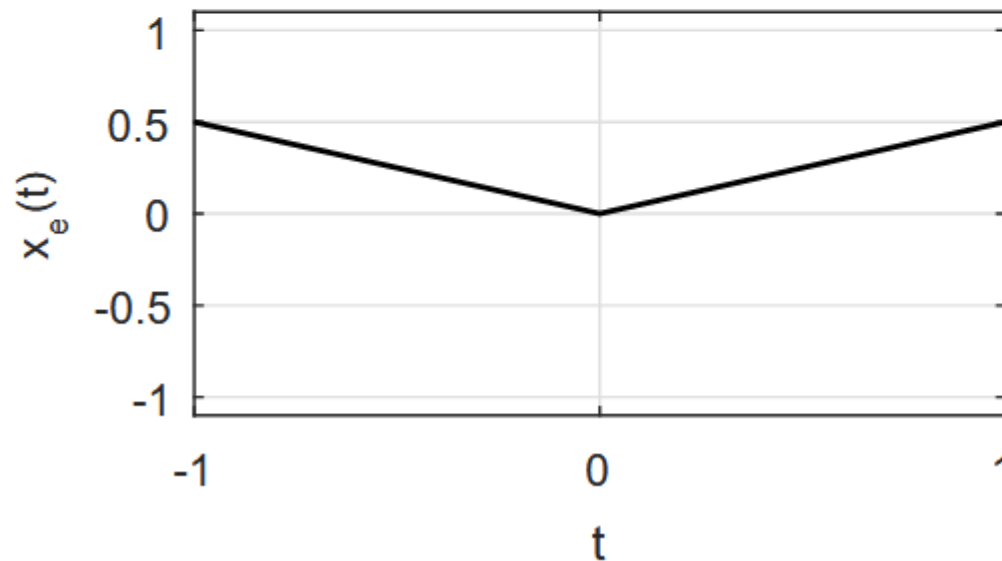
Solution

(b)

$$x_e(t) = 0.5[tu(t) - tu(-t)] = 0.5|t|$$

$$x_o(t) = 0.5[tu(t) + tu(-t)] = 0.5t$$

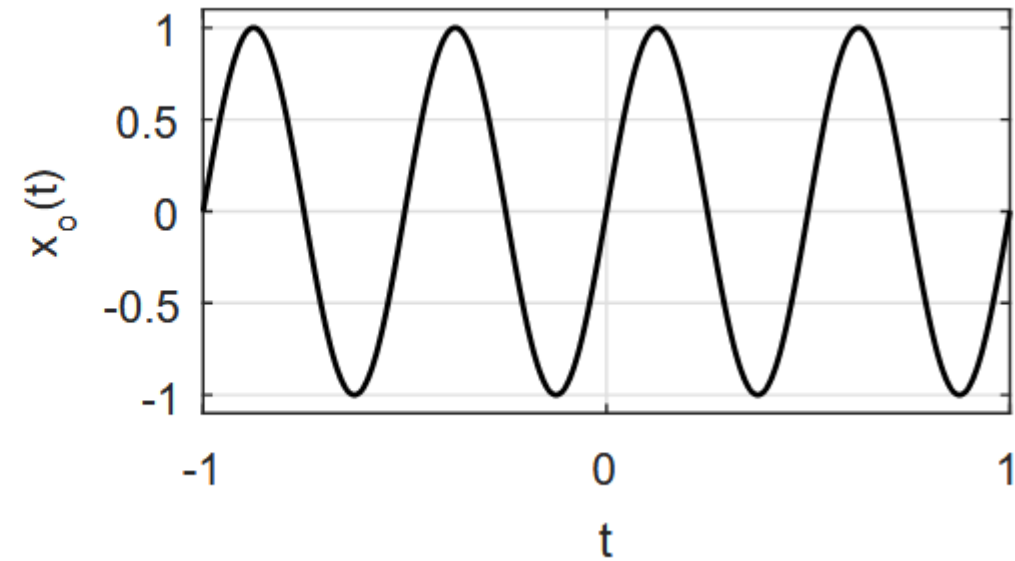
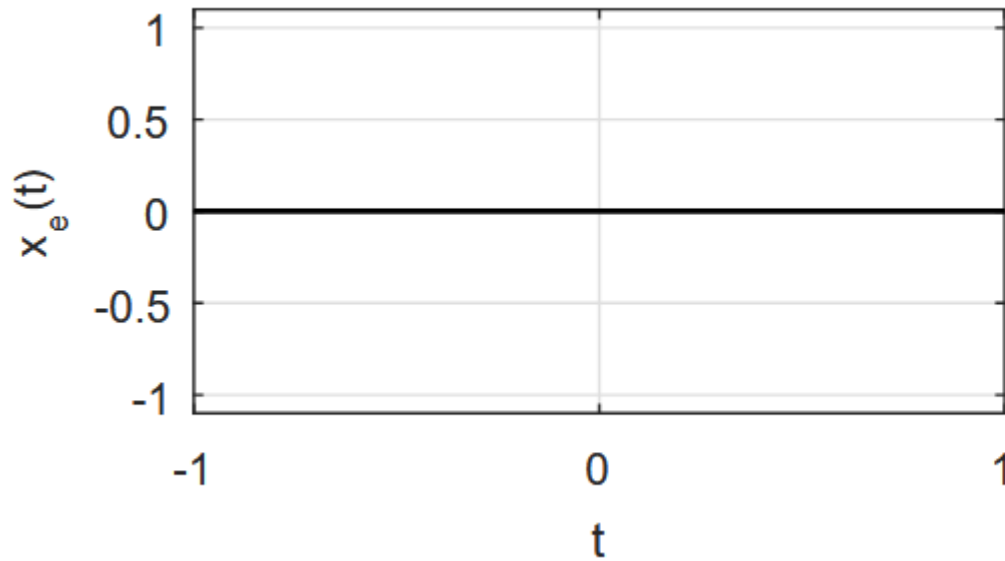
These component signals are shown in Fig. S1.5-1b.



Solution

$$x_e(t) = 0.5[\sin \omega_0 t + \sin(-\omega_0 t)] = 0$$

$$x_o(t) = 0.5[\sin \omega_0 t - \sin(-\omega_0 t)] = \sin \omega_0 t$$



Problem 6

Given $x_1(t) = \cos(t)$, $x_2(t) = \sin(\pi t)$, and $x_3(t) = x_1(t) + x_2(t)$.

Is $x_3(t)$ periodic?

Solution

For the sum of two signals to be periodic

- Both signals must be periodic
- Their periods must have an LCM (i.e., ratio of the periods should be rational).
 - If the above hold, the LCM is the period of the sum.

We know that $\cos(\omega t)$ and $\sin(\omega t)$ are periodic with $\frac{2\pi}{\omega}$

Therefore $T_1 = 2\pi$ and $T_2 = 2$

$$\frac{T_1}{T_2} = \pi \text{ which is not rational.}$$

Therefore $x_3(t)$ is not periodic.

Problem 7

Simplify the following expressions:

(a) $\left(\frac{\sin t}{t^2 + 2}\right)\delta(t)$

(b) $\left(\frac{j\omega + 2}{\omega^2 + 9}\right)\delta(\omega)$

(c) $[e^{-t} \cos(3t - 60^\circ)]\delta(t)$

(d) $\left(\frac{\sin\left[\frac{\pi}{2}(t - 2)\right]}{t^2 + 4}\right)\delta(1 - t)$

(e) $\left(\frac{1}{j\omega + 2}\right)\delta(\omega + 3)$

(f) $\left(\frac{\sin k\omega}{\omega}\right)\delta(\omega)$

Solution

Using the fact that $f(x)\delta(x) = f(0)\delta(x)$, we have

(a) 0

(b) $\frac{2}{9}\delta(\omega)$

(c) $\frac{1}{2}\delta(t)$

(d) $-\frac{1}{5}\delta(t - 1)$

(e) $\frac{1}{2-j3}\delta(\omega + 3)$

(f) $k\delta(\omega)$ (use L' Hôpital's rule)

Problem 8

Evaluate the following integrals:

(a) $\int_{-\infty}^{\infty} \delta(\tau) x(t - \tau) d\tau$

(b) $\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$

(c) $\int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$

(d) $\int_{-\infty}^{\infty} \delta(2t - 3) \sin \pi t dt$

(e) $\int_{-\infty}^{\infty} \delta(t + 3) e^{-t} dt$

(f) $\int_{-\infty}^{\infty} (t^3 + 4) \delta(1 - t) dt$

(g) $\int_{-\infty}^{\infty} x(2 - t) \delta(3 - t) dt$

Solution

In these problems remember that impulse $\delta(x)$ is located at $x = 0$. Thus, an impulse $\delta(t - \tau)$ is located at $\tau = t$, and so on.

(a) The impulse is located at $\tau = t$ and $x(\tau)$ at $\tau = t$ is $x(t)$. Therefore

$$\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau = x(t).$$

(b) The impulse $\delta(\tau)$ is at $\tau = 0$ and $x(t - \tau)$ at $\tau = 0$ is $x(t)$. Therefore

$$\int_{-\infty}^{\infty} \delta(\tau) x(t - \tau) d\tau = x(t).$$

Using similar arguments, we obtain

Solution

(c) 1

(d) 1

(e) e^3

(f) 5

(g) $x(-1)$

Problem 9

Find area under the given functions for $-2 < t < 3$

$$\delta(2t - 3) \sin \pi t dt$$

$$\delta(t + 3)e^{-t} dt$$

$$(t^3 + 4)\delta(1 - t) dt$$

Solution

$$\delta(2t - 3) \sin \pi t dt \longrightarrow \sin\left(\frac{3}{2}\pi\right) = -1$$

$$\delta(t + 3)e^{-t} dt \longrightarrow 0 \text{ (since impulse is not within region of integration)}$$

$$(t^3 + 4)\delta(1 - t) dt \longrightarrow 1^3 + 4 = 5$$

Problem 10

$$x(t) = De^{j\omega_0 t}$$

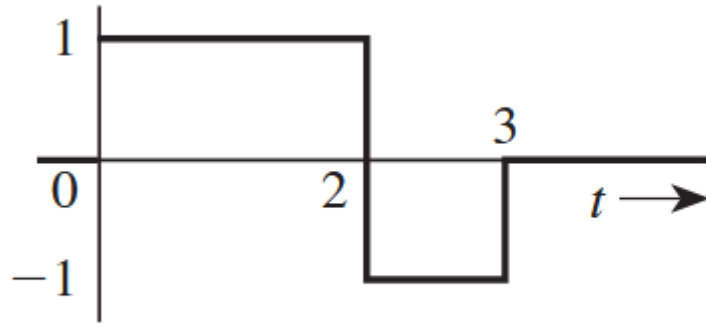
Is this a power or energy signal? Find both.

Solution

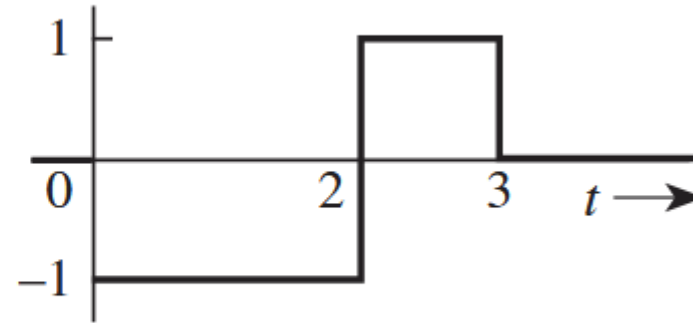
- This function is periodic with period $T = \frac{2\pi}{\omega_0}$
- Periodic signals have infinite energy and therefore cannot be energy signals.
- We can use power formula for periodic signal to integrate the function over one whole period and divide by the period T
- This would give $P_x = |D|^2$

Problem 11

Are the following signals energy signals or power signals? Find their energy and power.



(a)



(b)

Solution

$$\text{(a)} \quad E = \int_0^2 (1)^2 dt + \int_2^3 (-1)^2 dt = 3$$

$$\text{(b)} \quad E = \int_0^2 (-1)^2 dt + \int_2^3 (1)^2 dt = 3$$

- Both are energy signals as their energies are finite.
- Power of an energy signal is always zero.

Questions?? Thoughts??

